RTX PRO 6000 normalization comparison

This is an observation report for a software diagnostic panel. It does not certify a speedup, promote a research hypothesis, or establish an ECDLP advantage.

Run: RunPod pod `1lq0cd2l0en5ju`; full RTX PRO 6000 Blackwell Server Edition, 97,887 MiB; driver 595.91.07, CUDA 13.2.51, Rust 1.88.0. The pod was deleted after successful collection. No continuing GPU workload remains.

Source: `47ff700720f3563ed67eb1bb9e507db8c252c1c7`.

Protocol: EXPLICIT-RELATIONS-NORMALIZE-20261030, frozen before data. Four arms, three fixed instances, one warmup and nine measured repeats: 120 completed rows, 12 warmups and 108 measured. No failures or exclusions.

Exact work and validation

The product tree performs 765 source-level field multiplications per block in inversion preparation, versus 4608 for the old dual scan: a derived 83.3984375% reduction in this operation category. Both additionally pay one field inversion and the same affine conversions. This is not a total-GAE, machine-instruction or runtime reduction.

All variants matched independently checked CPU witnesses and normalization outputs. Host/CUDA service and bounded reduction tests, 10 Rust tests and Clippy passed. Memcheck: zero errors. Racecheck: zero hazards, errors and warnings. Synccheck: zero errors. Fixtures covered infinity, partial blocks and random projective scales including a 61-bit field.

Recorded normalization medians

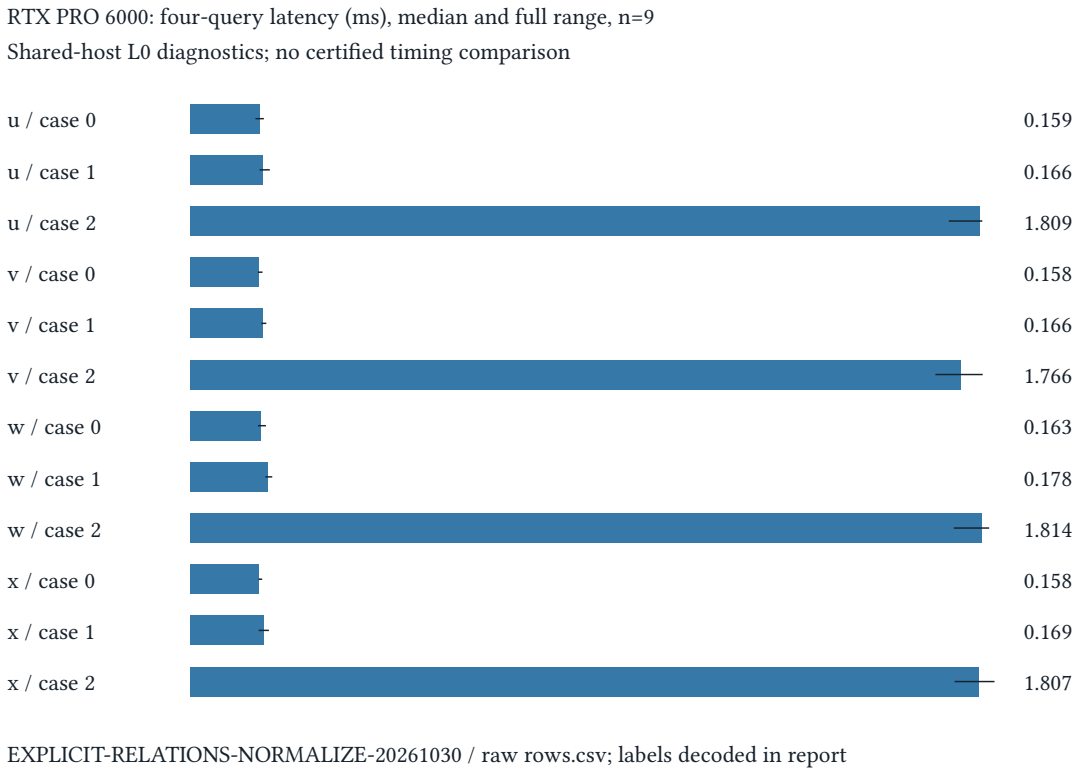
Milliseconds, n=9 per arm/cell. Kernel-only CUDA-event timing after warmup.

Lanes	Old scan	Tree	CUB	A/A scan
257	0.009620	0.009036	0.009576	0.009580
65,537	0.011332	0.010652	0.011168	0.011496
262,145	0.021668	0.018152	0.024024	0.021508

For four 256-point bases, four-query medians were 1.809353 ms old scan, 1.765703 ms tree, 1.814083 ms CUB and 1.807224 ms A/A. Old/tree query ranges overlap: 1.738424–1.815144 ms and 1.707625–1.815973 ms. Cold-pipeline medians were 143.981243 ms old scan and 145.227285 ms tree. The candidate did not have a lower cold median in this cell.

All clocks are L0 shared-host diagnostics; no end-to-end speedup is established.

Four-query diagnostics



Neutral labels: u=old scan, v=product tree, w=NVIDIA CUB BlockScan, x=second invocation of the exact u binary. Cases 0/1/2 have 16/64/256 points per base. Bars show medians with full observed ranges; n=9. Raw data and the frozen native analyzer are included in the same study directory.

Evidence and limits

The canonical checked record is `results/export-v2.json`. It corrects only an unobserved unmasking timestamp in the preserved earlier export; raw values, analyses and hashes are unchanged. The native analyzer consumed neutral labels, but no human-blinded inference is claimed. Planning time at freeze was an unmeasured allocation allowance; round 0 preserves actual preflight costs. No rows were removed for their timing.

The GPU normalization interval excludes fixture construction and readback. Prepare includes allocation, table build and sort. Queries include transfers, synchronization and native witness verification. Cold cost includes context startup, fixture construction, prepare and queries. The independent host comparison is validation overhead outside those intervals.

Raw archive: `s3://setec-astronomy-artifacts/validation/ecbench-explicit-x/runpod-1lq0cd2l0en5ju/`. SHA-256 of `results.tgz`: `d83cdcc92e49929fce81413ea1b2d9ebc807ff26093d2963c9735de6f1ff268b`. Download-back verification matched. No API credential is in the archive.

The source-level count, device tests and stage diagnostics do not establish full ECDLP speed, wider-field complete reduction, or a GPU tensor backend. Root crypto/ecbench integration is outside this run. Chunk synchronization and repeated right-side sums remain follow-up optimization targets.